

Carry your intermediate computations to at least four decimal places.

Q1- Apply three iterations of Newton Raphson method to the equation $-4x + \cos(x^2) + 2 = 0$, to estimate the root. Starting with the initial guess $x_0 = 0.5$.
(10 Marks)

Q3- Use the point (0,0) as the initial estimate of the *optimal gradient steepest method* to the multidimensional function

$$z = x^2 + y^2 + 2x + 4y \quad (10 \text{ Marks})$$

Perform one iteration to calculate the new values of x and y

(4 Marks)

Q2: a) Solve Equations $2x+5y=21$,

$$x+2y=8$$

using Cramer's Rule method

b) Perform two iterations to solve the following system of linear equations using the iterative Jacobi method. Make the initial guess $x_1 = 0, x_2 = 0, x_3 = 0$.
(6Marks)

$$8x_1 - 2x_2 + 3x_3 = -16$$

$$-3x_1 + 9x_2 + x_3 = 24$$

$$2x_1 - x_2 - 6x_3 = 36$$